ERBOT Results Summary

Synthetic 10K Dataset

ERBOT v0.2.0

Generated: 2026-03-23 | Run time: ≈11 minutes (SVM excluded)

Contents

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1 Dataset Overview

Property	Value
Dataset	Synthetic 10K (d10k)
Records	10,000
Fields	2 (`id`, `text`)
Task	Deduplication (single-source)
Ground truth	8,705 matched pairs (from `10Kduplicates.csv`)

The synthetic 10K dataset consists of 10,000 noisy text records generated to simulate real-world deduplication challenges. Each record has a single free-text field (`text`) containing a scrambled mix of tokens — names, dates, numbers, addresses — with no explicit field structure. Duplicates are introduced by corrupting original records with character substitutions, token deletions, and reorderings, mimicking common data entry and OCR errors.

Stage	Setting	Value
Blocking	Method	Prefix blocking on `text`, length 3
Blocking	Candidate pairs	885,301
Blocking	Reduction ratio	0.982 (98.2% of all pairs eliminated)
Similarity	Fields computed	1 (`text` only)
Weights	Method	Equal (weight = 1.0 for `text`)
Clustering	Methods run	8 (SVM excluded — see note)
Merge	Strategy	Best

2 Pipeline Configuration

SVM excluded: SVM was initially run but terminated after ≈20 hours without completing. With 885,301 candidate pairs, SVM training time was estimated at 1–4 days due to its super-linear scaling with the number of pairs. All other 8 methods completed in ≈11 minutes. SVM remains viable for smaller datasets (CORA at 137K pairs completed in ≈63 minutes total) but is not practical at this scale without subsampling or approximation. A scalable alternative — logistic regression or gradient boosting — is recommended as a replacement.

Blocking note: 885,301 pairs from 10,000 records (49,995,000 total possible). Prefix-3 achieves 98.2% reduction — the highest across all three datasets tested. However, for a single noisy text field, many true matches will have corrupted or reordered opening tokens and are silently discarded at this stage.

3 Performance Results

8 clustering methods evaluated (SVM excluded):

Method	ARI	NMI	VI	B^3-P	B^3-R	B^3-F	Hom.	Com.	V
leiden	0.000	**0.914**	**1.97**	**1.000**	0.271	**0.426**	**1.000**	0.841	**0.914**
threshold_cc	**0.023**	0.778	4.21	0.334	0.519	0.406	0.707	0.865	0.778
louvain	**0.023**	0.778	4.21	0.334	0.519	0.406	0.707	0.865	0.778
label_prop	**0.023**	0.778	4.21	0.334	0.519	0.406	0.707	0.865	0.778
hclust_avg	0.002	0.371	8.82	0.019	0.573	0.036	0.249	0.727	0.371
pam	0.002	0.371	8.82	0.019	0.573	0.036	0.249	0.727	0.371
hclust_ward	0.001	0.347	8.80	0.019	0.690	0.037	0.224	0.772	0.347
gc	0.000	0.000	10.4	0.001	1.000	0.002	0.000	1.000	0.000
final	0.023	0.778	4.21	0.334	0.519	0.406	0.707	0.865	0.778

Metrics: ARI = Adjusted Rand Index; NMI = Normalized Mutual Information; VI = Variation of Information (lower is better); B^3 = B-cubed; V = V-measure; Hom. = Homogeneity; Com. = Completeness. Full PairF values in `er_performance.csv`.

4 Final Cluster Assignment

The `final` strategy correctly selected **threshold_cc** (highest ARI = 0.023).

Statistic	Value
Total clusters	1,619
Records	10,000
Mean cluster size	6.2

5 Method-by-Method Interpretation

Threshold-CC / Louvain / Label Propagation (Highest ARI — 0.023)

The three graph-based methods again converge to identical solutions, confirming that similarity graph structure dominates algorithm choice. B^3-F of 0.406 with precision 0.334 and recall 0.519 is a moderate result — about half of true matches are found at the cost of some false merges. These are the best practical methods at this scale.

Leiden (ARI = 0.000, NMI = 0.914, B^3-F = 0.426)

Leiden produces the highest NMI and B^3-F but zero ARI. This apparent contradiction arises because ARI heavily penalises over-splitting: Leiden creates many small, highly pure sub-clusters (B^3-precision = 1.000) but each true entity is fragmented across multiple predicted clusters (recall = 0.271). The clusters are perfectly pure but incomplete.

For applications where **false merges are very costly** (e.g. legal identity resolution where merging two different people is a serious error), Leiden at its default resolution is actually the safest choice. For applications where **recall matters**, threshold-CC is better.

hclust_avg / PAM (ARI ≈ 0.002)

Near-zero ARI despite moderate NMI. Both achieve near-zero precision (0.019), meaning almost every pair they merge is a false positive. At 10K records the similarity matrix is too sparse and noisy for centroidal/hierarchical methods to find meaningful structure.

hclust_ward (ARI = 0.001)

Similar to hclust_avg. Ward linkage performs marginally worse on precision but slightly better on recall. Hierarchical methods are clearly not suited to large sparse similarity matrices.

GC — Graph Coloring (Degenerate — ARI = 0.000)

Perfect recall, near-zero precision, collapsed to one giant cluster. The gc dimension mismatch error persists across all three datasets — this is a recurring bug in the GCMER interface layer that needs fixing.

SVM (Not Run)

SVM was launched but killed after ≈20 hours still processing 885,301 pairs. Based on observed runtimes on smaller datasets (CORA 137K pairs: included in 63 min; affiliation 164K pairs: dominated a 5.6-hour run), estimated completion time for 885K pairs was 1–4 days. SVM's kernel matrix computation scales super-linearly with the number of training examples, making it impractical at this dataset size. Logistic regression or gradient boosting are recommended as drop-in replacements with near-linear scaling.

6 Key Findings

1. **Overall performance is very low.** Best ARI is 0.023 — lower than affiliation (0.076) and far below CORA (0.746). At 10,000 records with a single noisy unstructured text field, the task is fundamentally harder.
2. **Blocking discards too many true matches.** 98.2% reduction is aggressive. With only 8,705 gold pairs in a 10,000-record dataset, many duplicates involve records with corrupted or reordered first tokens, which prefix-3 blocking eliminates silently. Blocking is the primary performance bottleneck.
3. **Leiden is the precision champion.** Perfect B^3-precision (1.000) and NMI (0.914) make Leiden the best choice when false merges must be avoided. Its ARI = 0 reflects over-splitting, not noise.
4. **SVM is not viable at this scale.** For datasets above ≈200K candidate pairs, SVM should be replaced with a faster supervised classifier.
5. **Graph methods plateau quickly.** Threshold-CC, Louvain, and Label Propagation give identical results here as in the previous two datasets. The similarity graph structure consistently dominates community detection algorithm choice.
6. **The `final` merge correctly selected threshold_cc.** Merge bug fix continues to work correctly across all three datasets.

7 Recommendations for Next Steps

Priority	Action
High	Replace prefix-3 blocking with SN blocking (window = 40–80) — the 98.2% reduction is too aggressive for a noisy single-field dataset.
High	Replace SVM with logistic regression or gradient boosting for supervised classification — near-linear scalability, similar accuracy.
Medium	Tune Leiden resolution downward (try 0.1–0.3) to improve recall while preserving its high precision.
Medium	Fix GC dimension mismatch error — silently falls back to degenerate solution across all datasets.
Medium	Measure pair completeness at Stage 3 to quantify how many gold pairs are lost at blocking.
Low	Explore token-set / Jaccard similarity instead of default string similarity for unstructured text.
Low	Try field parsing: split `text` into sub-fields (name, date, address tokens) to enable multi-field similarity.

8 Cross-Dataset Comparison

The clear trend: **more records + fewer fields = lower performance**. CORA benefits from 15 discriminative fields that collectively identify duplicates even under noise. Affiliation and Syn10k each have one field, forcing all discriminative power through a single noisy similarity score.

9 Output Files

Metric	CORA	Affiliation	Syn10k
Records	1,879	2,260	10,000
Fields (usable)	15	1	1
Gold pairs	64,578	32,816	8,705
Candidate pairs	137,707	164,719	885,301
Blocking RR	92.2%	93.5%	98.2%
Best ARI	0.746	0.076	0.023
Best B^3-F	0.738	0.460	0.426
Best method (ARI)	hclust_avg	threshold_cc	threshold_cc
SVM viable?	Yes (≈63 min)	Marginal (≈5.6h)	No (≈1–4 days)
Final correct?	Yes	Yes	Yes

File	Description
`er_performance.csv`	Full metric table (9 methods × 13 metrics)
`er_predictions.csv`	Final cluster assignments (10,000 records × 2 columns)
`d10k_data_profile.pdf`	Dataset profile: field types, distributions, missingness
`plot_method_comparison.pdf`	Grouped bar chart: ARI, B^3-F, NMI per method
`plot_performance_heatmap.pdf`	Colour heatmap: all metrics × all methods
`plot_cluster_sizes.pdf`	Cluster size distribution histogram
`plot_precision_recall.pdf`	Precision vs. Recall scatter (B-cubed and Pair-F)
`SYN10K_RESULTS_SUMMARY.pdf`	This document